Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Software Classification

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Issue	Rev.	Date	Change
1	0	2026-10-10	First draft
1	1	2026-10-11	Safety-critical designation from the hazard analysis

1.0 Purpose and Scope

This document establishes the software classification of each software item, and so the rigour each item requires. It is the formal record of the software classification determinations [1, Sec. 3.5.2, p. 25].

The evaluated software items encompass the System software and its enabling systems, as defined in the Concept of Operations [4, Sec. 1.2.2]. The System models an uncrewed spacecraft whose primary mission objectives directly depend on this software; a failure would cost those objectives.

2.0 Documents

[1] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[2] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022.
[3] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, page "SWE-020 – Software Classification".
[4] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 0, Draft.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Hazard Analysis*, Issue 1, Revision 0, Draft.

3.0 Classification

Each item takes the highest class whose definition applies [1, Sec. 3.5.1, p. 24].

Software item	Role	Class
Controller firmware	FLIGHT	B
Onboard computer software	FLIGHT	B
Ground software in the mission operations center	GROUND	B
Bench test tools	TEST	C
Firmware build tools	BUILD	D

Legend: role

FLIGHT	Operates the spacecraft from orbit.
GROUND	Operates the spacecraft from the ground.
TEST	Tests the System; never operates the spacecraft.
BUILD	Builds the System; never operates the spacecraft.
Class	As defined in [1, App. D, pp. 79–86].

Note. *The class decides which rules of NPR 7150.2D apply to each item [1, Sec. 3.1.11, p. 22]: all 100 rules apply to Class B, 92 to Class C and 64 to Class D [1, App. C, pp. 56–78]. Appendix B records which of them the project follows.*

3.1 Rationale

Controller firmware: Class B. Class B covers flight software "required to operate the vehicle or space asset ... such as commanding of the vehicle or asset" [1, App. D, p. 81]; its examples include "fault protection" [1, App. D, p. 81]. The controller runs commands and returns to working firmware on its own when new firmware fails.

Onboard computer software: Class B. The same definition applies. It passes every command and firmware image to the controller, watches it, and restores it when its firmware cannot run.

Ground software: Class B. Class B examples include "command and control ground systems" [1, App. D, p. 81]. The ground software prepares and checks every command and firmware image sent to the spacecraft.

Bench test tools: Class C. Class C covers "software used for the testing of space assets" [1, App. D, p. 82], and Class B excludes systems "used in testing Class B systems" in a development environment [1, App. D, p. 81].

Firmware build tools: Class D. Class D covers "ground software tools that support engineering development" [1, App. D, p. 83].

4.0 Safety-Critical Designation

Some software is safety-critical: the parts that can cause a hazard, or that exist to prevent, detect or handle one [2, Sec. 4.2, p. 20]. Safety-critical software must meet extra requirements on top of its class, for example test coverage of every decision (MC/DC) [1, Sec. 3.7.4, pp. 26–27].

The hazard analysis designates the items below from its hazards [5, Sec. 4.0]. Until the architecture divides each item into components, the whole item is treated as safety-critical.

Software item	Safety-critical	Criteria	Hazards
Controller firmware	YES	a, b, c, e	H-1 to H-4, H-6 to H-13
Onboard computer software	YES	a, b, c, e	H-1 to H-17
Ground software	YES	a, b	H-2, H-4 to H-7, H-9, H-10, H-15, H-17
Bench test tools	NO	NONE	NONE
Firmware build tools	NO	NONE	NONE

Legend: criteria [2, Sec. 4.2, p. 20]. a: causes or contributes to a system hazardous condition or event. b: controls functions identified in a system hazard. c: provides mitigation for a system hazardous condition or event. d: mitigates damage if a hazardous condition or event occurs. e: detects, reports and takes corrective action if the system reaches a potentially hazardous state. Hazards: the hazard IDs of [5, Sec. 4.0].

Note. *Bench test tools stay out of operation (H-12). Firmware build tools never run in operation; a defect they put into firmware is a shared cause [5, Sec. 5.2].*

5.0 Tailoring and Review

- **Joint agreement and single-developer tailoring:** Software engineering and software assurance are expected to agree on each classification [1, Sec. 3.5.1 Note, p. 25]. One engineer develops the System, with no independent software assurance team (ConOps constraint C-3); the classification therefore rests with the author alone.
- **Life-cycle re-evaluation:** The classification is revisited at each later phase [3, Sec. 3.5.1]. If an item's class changes, the plans are updated to match [1, Sec. 2.2.8, p. 19].

A Open Items

The values below are not yet known.
TBD-1 Project name and document reference.
The author accepted the safety-critical designation (Section 4.0) on 11 October 2026.

B Requirements Mapping Matrix

NPR 7150.2D sets 100 rules for how software is built, each marked for the classes it applies to [1, App. C, pp. 56–78]. The project intends to follow every rule that applies to an item's class, except the 20 tailored out in the second table below [1, Sec. 3.1.12, p. 22]. The author decided this tailoring on 10 October 2026; it is revisited at each later phase.

Class	Rules that apply	Tailored out	Followed
B	100	20	80
C	92	15	77
D	64	11	53

Followed by the Class B items (by NASA's rule number, SWE): 013, 016, 017, 018, 020, 023, 024, 027, 033, 034, 036, 037, 050, 051, 052, 053, 054, 055, 057, 058, 060, 061, 062, 063, 065, 066, 068, 071, 073, 075, 077, 079, 080, 081, 082, 083, 084, 085, 086, 087, 088, 089, 090, 093, 121, 125, 134, 135, 136, 139, 146, 147, 154, 156, 157, 159, 176, 184, 185, 186, 187, 189, 190, 191, 192, 193, 194, 195, 196, 199, 200, 201, 202, 203, 204, 207, 210, 211, 219, 220. The Class C and D items follow those of these rules marked for their class.

SWE	Requirement	Reason
141	Software IV&V for NASA project categories	NOT NASA
143	Architecture review for NASA project categories	NOT NASA
174	Planning data to the Center measurement repository	NOT NASA
148	Reuse candidates to NASA's catalogue	NOT NASA
094	Measurement data to the sponsoring organisation	NOT NASA
039	Status reports from the software developer	NO SUPPLIER
040	Products and records from the developer to NASA	NO SUPPLIER
042	NASA access to the developer's source code	NO SUPPLIER
046	Software schedule from the developer	NO SUPPLIER
206	NASA access to the developer's models and simulations	NO SUPPLIER
045	Joint NASA/developer audits	NO SUPPLIER
131	IV&V plan	NO IV&V
178	IV&V access to development artifacts	NO IV&V
179	Responses to IV&V issues and risks	NO IV&V
032	CMMI-DEV rated organisation	NO RATING
022	Software assurance and safety plan per NASA-STD-8739.8	NO ASSURANCE
205	Safety-critical determination with the SMA organisation	NO ASSURANCE
015	Two cost estimate models	NO BUDGET
151	Conditions on the cost estimate	NO BUDGET
070	Accredited models and simulations for flight qualification	BENCH ONLY

Legend: reason

NOT NASA The System is not a NASA program or project; there is no NASA Center, catalogue or sponsoring organisation.

NO SUPPLIER The engineer develops all software; there is no separate developer or supplier.

NO IV&V Independent verification and validation is not performed.

NO RATING There is no organisation to hold a CMMI rating.

NO ASSURANCE There is no independent software assurance or SMA organisation (ConOps constraint C-3); the author decides alone.

NO BUDGET The project has no development budget to estimate.

BENCH ONLY The System is demonstrated on a bench, not qualified for flight (ConOps constraint C-4).